

Colour filters are often used in front of the lens to selectively block some colours while passing others (similar to how colour filters are used for each pixel in a digital camera's bayer array). Filters are named after the hue of the colour which they pass, not the colour they block. These can block all but a primary colour such as red, green or blue, or can partially block any weighted combination of the primary colours (such as orange or yellow). Careful selection of these filters allows the photographer to decide which colours will produce the brightest or darkest tones.



There's a world of difference between using plain old Greyscale and tweaking a Black-and-white conversion

Converting a digital colour photo into black and white utilises the same principles as with colour filters in film photography, except filters instead apply to each of the three RGB colour channels in a digital image. Whether you specify it or not, all conversion techniques have to use some weighted combination of each colour channel to produce greyscale brightness. Some techniques assume a combination for you, although the more powerful ones give you full control. Each makes its own trade-offs between power and ease of use, and so you may find that some techniques are best suited only to certain tasks.

The channel mixer tool in Photoshop allows the user to control how much each of the three colour channels (red, green and blue) contribute to the final greyscale brightness. It is undoubtedly one of the most powerful black and white conversion methods.